Projectivity of $A(G)$ over its standard Segal algebras forces compactness

Solution packet for the open question in arXiv:0907.0986

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Full positive answer. For $S = S^1A(G)$ or $S = S_0(G)$, the Fourier algebra $A(G)$ is operator projective as a left S-module if and only if G is compact. The converse reduces to a general range observation: a projective splitting $A(G) \to S\widehat{\otimes}A(G)$, followed by multiplication, takes values in S and is the identity in $A(G)$; hence $S = A(G)$. Since both choices of S embed in $L^1(G)$, this would continuously embed $A(G)$ in $L^1(G)$. Disjoint translates of one compactly supported regular coefficient have L^1 norm of order n but $A(G)$ norm of order at most $\sqrt{n}$, ruling out noncompact G.

Source question

Forrest, Lee, and Samei study operator projectivity of natural modules over the Lebesgue–Fourier algebra $S^1A(G)$ and Feichtinger's algebra $S_0(G)$ [1]. Both are operator Segal algebras of $A(G)$. The source proves projectivity of $A(G)$ over either algebra when G is compact and asks in its introduction, on source PDF page 2, whether the converse holds:

> "...it remains open whether or not the projectivity of $A(G)$ as either an $S^1A(G)$ or an $S_0(G)$-module forces G to be compact."

as well as that of the underlying group G. (See [11] and [24] for a more detailed account of theses algebras.)

As we did in [9], we will show that discreteness plays an imprortant role in identifying projective modules for both $S^1A(G)$ and $S_0(G)$. This time, we also see that compactness plays an equal role. As such we find that many natural spaces are projective when viewed as modules over $S^1A(G)$ or $S_0(G)$ if and only if G is finite. There are exceptions to this rule noteably $A(G)$, which is projective when G is compact, and $L^1(G)$, which is projective if and only if G is discrete. This latter result reflects the natural duality between $A(G)$ and $L^1(G)$, though at this point it remains open whether or not the projectivity of $A(G)$ as either an $S^1A(G)$ or an $S_0(G)$-module forces G to be compact.

In the latter part of the paper, we will change directions once more. This time instead of looking at an operator Segal algebra of $A(G)$, we will instead focus our attention on another algebra $A_{cb}(G)$ in which $A(G)$ is itself the operator Segal

Proof intuition

The splitting criterion for an essential projective module normally appears to say only that multiplication lands back in the ambient algebra. Here it says more. If S is a Segal ideal in a commutative Banach algebra A, then multiplication

$$S\widehat{\otimes}A \longrightarrow S$$

is bounded in the stronger Segal norm. Consequently, any right inverse $A \to S\widehat{\otimes}A$ to multiplication forces every element of A to lie in S. Thus a proper essential Segal ideal cannot have its ambient algebra as a projective module.

For the two source algebras, equality $S = A(G)$ forces every Fourier-algebra function to be integrable. On a noncompact group, however, one may place arbitrarily many disjoint translates of a fixed compactly supported regular coefficient. Disjointness makes their L^1 norms add, whereas the coefficient realization in the left regular representation makes their $A(G)$ norm grow only like the Hilbert norm of an orthogonal sum.

The abstract range lemma

Lemma 1. *Let A be a commutative completely contractive Banach algebra and let S be an operator Segal algebra in A in the sense of [1]: S is dense in A, is an ideal, is a completely contractive A-module in its own norm, and AS is dense in S in that norm. If A is operator projective as a left S-module, then $S = A$ as sets and the two norms are equivalent.*

Proof. First A is an essential S-module. Indeed, AS is dense in S in the S-norm. The inclusion $j : S \to A$ is continuous, so

$$S \subseteq \overline{AS}^{\|\cdot\|_A}.$$

Since S is dense in A, it follows that $\overline{SA}^{\|\cdot\|_A} = A$.

The essential-module projectivity criterion [1, Proposition 2.5] therefore supplies a completely bounded S-module map

$$\rho : A \longrightarrow S\widehat{\otimes}A$$

such that $\pi_A\rho = \mathrm{id}_A$, where $\pi_A(s \otimes a) = sa$ is viewed as taking values in A. But the Segal-module axiom and commutativity give a completely bounded map

$$\pi_S : S\widehat{\otimes}A \longrightarrow S, \qquad \pi_S(s \otimes a) = sa,$$

and $j\pi_S = \pi_A$. Hence the bounded map $J = \pi_S\rho : A \to S$ satisfies

$$jJ = \mathrm{id}_A .$$

Because j is the set-theoretic inclusion, every $a \in A$ equals the element $J(a) \in S$. Thus $A = S$, and boundedness of J together with boundedness of j makes their norms equivalent. □

The integrability obstruction

Lemma 2. *If the inclusion $A(G) \hookrightarrow L^1(G)$ is well-defined and bounded for a locally compact group G, then G is compact.*

Proof. Assume that G is noncompact and

$$\|v\|_1 \le C\|v\|_{A(G)} \qquad (v \in A(G)). \tag{1}$$

Choose a relatively compact measurable neighborhood V of the identity with positive Haar measure, put $\xi = 1_V \in L^2(G)$, and define

$$u(s) = \langle \lambda(s)\xi, \xi\rangle.$$

Then $u \in A(G)$ is nonzero, nonnegative, continuous, and supported in the compact set VV^{-1}; in particular $0 < \|u\|_1 < \infty$.

Choose a compact set K containing $V \cup VV^{-1}$. Noncompactness permits, for every n, elements $x_1, \dots, x_n$ whose left translates x_iK are pairwise disjoint: inductively choose x_{m+1} outside the finite union $\bigcup_{i\le m} x_iKK^{-1}$. Put

$$u_i(s) = u(x_i^{-1}s).$$

Their supports are disjoint, so

$$\left\|\sum_{i=1}^{n} u_i\right\|_1 = n\|u\|_1. \tag{2}$$

On the other hand,

$$u_i(s) = \langle \lambda(s)\xi, \lambda(x_i)\xi \rangle,$$

and the vectors $\lambda(x_i)\xi$ are orthogonal because the sets x_iV are disjoint. The coefficient definition of the Fourier-algebra norm yields

$$\left\|\sum_{i=1}^{n} u_i\right\|_{A(G)} \leq \|\xi\|_2 \left\|\sum_{i=1}^{n} \lambda(x_i)\xi\right\|_2 = \sqrt{n}\,\|\xi\|_2^2. \tag{3}$$

Combining (1)–(3) gives $n\|u\|_1 \leq C\sqrt{n}\,\|\xi\|_2^2$ for every n, a contradiction. □

Full answer

Theorem 1. *Let G be a locally compact group and let $S = S^1A(G)$ or $S = S_0(G)$. Then $A(G)$ is operator projective as a left S-module if and only if G is compact.*

Proof. Suppose first that $A(G)$ is operator projective over S. Lemma 1 gives $A(G) = S$ with equivalent norms. For $S = S^1A(G)$, the definition $S^1A(G) = A(G) \cap L^1(G)$ immediately gives a bounded inclusion $A(G) \hookrightarrow L^1(G)$. For $S = S_0(G)$, the source recalls that $S_0(G)$ is also an operator Segal algebra of $L^1(G)$, so its inclusion into $L^1(G)$ is bounded; equality $A(G) = S_0(G)$ gives the same conclusion. Lemma 2 implies that G is compact.

Conversely, the compact case is the positive result already recorded in [1]: $A(G)$ is operator projective over both $S^1A(G)$ and $S_0(G)$. □

Verification, scope, and novelty check

The proof uses only the source's definitions and projectivity criterion, the standard coefficient description of $A(G)$, and elementary Haar-measure geometry. It proves the exact converse for both named Segal algebras and, abstractly, shows that an ambient commutative algebra cannot be an essential projective module over a proper operator Segal ideal.

On 11 August 2026, the four run indexes and bounded exact-title/exact-phrase searches were checked for '0907.0986', the quoted open sentence, projectivity of $A(G)$ over $S^1A(G)$ or $S_0(G)$, and the general proper-Segal-ideal lemma. No direct answer was located. A 2017 paper entitled *Homological properties of Banach modules over abstract Segal algebras* cites the source and is a material novelty risk, but its searchable abstract does not state this operator-module theorem and full text was not available in the bounded check. Novelty is therefore provisional; the proof itself is self-contained.

Human-review recommendation

Review as a likely valid full positive solution. Verify especially that the source's operator Segal axioms give the bounded map $S\widehat{\otimes}A(G) \to S$ and that essentiality passes from AS dense in S to SA dense in A. The remaining compactness obstruction is an explicit norm-growth contradiction.

References

[1] B. E. Forrest, H. H. Lee, and E. Samei, *Projectivity of modules over Segal algebras*, arXiv:0907.0986; Houston J. Math. **39** (2013), 531–560.

[2] R. Nasr-Isfahani, M. Nemati, and S. Soltani Renani, *Homological properties of Banach modules over abstract Segal algebras*, Math. Slovaca **67** (2017), 191–198.